Chapter 5

The Need for a New Theory: General Relativity vs. Quantum Mechanics

O

ur understanding of the physical universe has deepened profoundly during the past century. The theoretical tools of quantum mechanics and general relativity allow us to understand and make testable predictions about physical happenings from the atomic and subatomic realms all the way through phenomena occurring on the scales of galaxies, clusters of galaxies, and beyond to the structure of the whole universe itself. This is a monumental achievement. It is truly inspiring that beings confined to one planet orbiting a run-of-the-mill star in the far edges of a fairly ordinary galaxy have been able, through thought and experiment, to ascertain and comprehend some of the most mysterious characteristics of the physical universe. Nevertheless, physicists by their nature will not be satisfied until they feel that the deepest and most fundamental understanding of the universe has been unveiled. This is what Stephen Hawking has alluded to as a first step toward knowing “the mind of God.”1

There is ample evidence that quantum mechanics and general relativity do not provide this deepest level of understanding. Since their usual domains of applicability are so different, most situations require the use of quantum mechanics or general relativity, but not both. Under certain extreme conditions, however, where things are very massive and very small—near the central point of black holes or the whole universe at the moment of the big bang, to name two examples—we require both general relativity and quantum mechanics for proper understanding. But like the mixing of fire and gunpowder, when we try to combine quantum mechanics and general relativity, their union brings violent catastrophe. Well-formulated physical problems elicit nonsensical answers when the equations of both these theories are commingled. The nonsense often takes the form of a prediction that the quantum-mechanical probability for some process is not 20 percent or 73 percent or 91 percent but infinity. What in the world does a probability greater than one mean, let alone one that is infinite? We are forced to conclude that there is something seriously